

SAMKELO SITHEMBISO MNGOMA

SOFTWARE & DATA ENGINEERING | CLOUD DELIVERY | TECHNICAL OPERATIONS

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Portfolio: mngomazar.github.io/samkelo-mngoma-portfolio

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PROFESSIONAL PROFILE

Hands-on software and data practitioner with experience in database administration, analytics, technical operations and cloud delivery. Co-founder and COO of SAICTS, combining stakeholder coordination with implementation across mobile applications, learning platforms and data systems. Seeking software, data, cloud delivery or technical operations opportunities.

CORE SKILLS

Software: Python, TypeScript, JavaScript, React Native, FastAPI, Node.js, REST APIs.

Data: SQL, PostgreSQL, Power BI, data cleaning, transformation, modeling and reporting.

Cloud & delivery: AWS Lambda/API Gateway/Amplify/CloudWatch, Oracle Cloud, Docker, GitHub Actions, Linux, Expo/EAS and integration testing.

Operations: Requirements, documentation, stakeholder communication, role-based access, secret hygiene and release coordination.

PROFESSIONAL EXPERIENCE

Co-founder & Chief Operating Officer

SAICTS | Mar 2024 - present

- Coordinate project delivery, operational processes, resources, risks and stakeholder communication.
- Contribute software, API, data and cloud implementation across institutional elections, USSD services, learning and marketplace platforms.
- Maintain deployment documentation, test evidence and release checks across GitHub, Oracle and Expo delivery workflows.

Data Specialist (contract)

Lindbong Development | Jan 2023 - Mar 2024

- Cleaned, transformed and analyzed data; prepared visualizations and reports to support business decisions.
- Translated analysis into stakeholder-facing insights and coordinated technical requirements.

Database Administrator (contract)

Colman Protection | Aug - Dec 2023

- Worked on database setup, cloud configuration, access control, backup/recovery and integration planning.

Customer Retention & Data Operations

AA Wealth (insurance) | 2020 - 2023

- Supported customer retention, data processing and cleaning, financial-data review and business communication.

LEADERSHIP

Esayidi TVET SRC President (2021/2022) and Secretary General (2022/2023): student representation, governance, administration and stakeholder coordination.

SELECTED PROJECTS & DEVELOPMENT

SELECTED ENGINEERING & ANALYTICS WORK

MUT SRC elections - voter verification & operations

SAICTS institutional delivery | 2025

- Prepared voter-roll imports covering 12,000+ records, with cleaning, duplicate handling and database integrity checks.
- Worked on barcode verification, administrative controls, activity logs and reporting using AWS Lambda, API Gateway, Amplify and CloudWatch.

Elangeni TVET SRC elections & USSD services

SAICTS institutional delivery | 2025

- Contributed USSD/web voting workflows, voter eligibility checks, administrative monitoring and campus results/reporting.
- Co-authored the August 2025 election report; analyzed USSD sessions, retries, menu interactions and time-based billing reconciliation.

Madabukela Academy - integrated learning platform

SAICTS technical delivery | 2026 | live public platform

- Worked on assessments, evidence portfolios, offline queues, immediate feedback and context-aware tutor routing for tablet learners.
- Node.js/PostgreSQL/Docker on Oracle Cloud; verified public desktop/tablet flows. madabukela-academy.co.za

Papzi - creator marketplace

SAICTS contracted engineering | 2026 | deployed TestFlight beta

- Worked across Expo/React Native, FastAPI/PostgreSQL, role access, scheduled bookings, messaging, road routing and release configuration.
- Delivered Oracle Docker/GHCR backend and TestFlight beta; CI verified 300 concurrent read-test sessions and 3,000 requests without errors.

Commercial performance reporting

Independent Power BI assignment | 2026

- Built native multi-page sales reporting with executive, regional, sales-manager, brand-trend and customer views, interactive filters and recommendations. Independent assignment; private source data is not published.

Drone telemetry & Phepha safety intelligence

Engineering prototypes | 2026

- ESP32/Raspberry Pi: inertial/distance sensing, receiver decoding, JSON serial telemetry, CSV/web monitoring and heartbeat/arming failsafes. Bench prototypes, not flight-certified systems.
- Phepha: Express/WebSocket API aggregation, MapLibre layers, TTL caching and source-attributed public-data integration.

EDUCATION & PROFESSIONAL DEVELOPMENT

NCV Finance, Economics & Accounting, Levels 2-4

Esayidi TVET College | 2020-2022

Online professional development: IBM Applied Data Science, data engineering, DevOps, CI/CD and application security; Google Data Analytics; Microsoft/freeCodeCamp Foundational C#; Alison database systems, networking and cybersecurity (CPD). Certificate copies available on request.

Forage virtual job simulations (2023): KPMG Data Analytics and JPMorgan Chase analyst/private-banking exercises. Training, not employment.